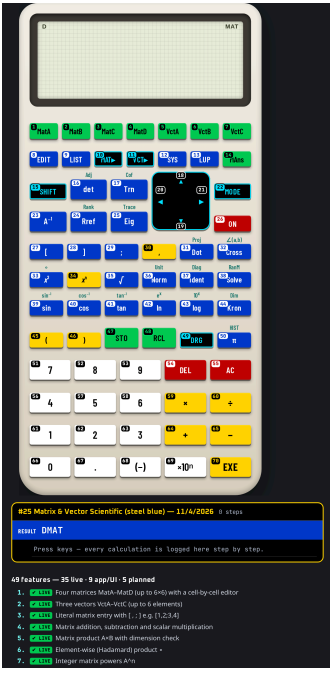
U-45 CALCULATORS

User Manual

Calculator #25 Matrix & Vector Scientific (steel blue)

Scientific – natural display



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#25 — Matrix & Vector Scientific (steel blue)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 MatA]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

algebraic matrix entry (MatA…MatD, VctA…VctC or [1,2;3,4], then EXE).

Chapters

1. Getting started
2. Features
3. Feature guide (2b)
4. Numbered key map
5. Key-by-key reference
6. Worked examples
7. Tricks & hacks
8. Keyboard shortcuts
9. Common mistakes
10. Practice drills
11. Appendix A — Drill answers

12. Appendix B — Function finder (A–Z)
13. Appendix C — Glossary
14. Appendix D — Model notes & limits
15. Appendix E — Quick-reference card

1 · Getting started

Open U-45 CALCULATORS, choose **#25** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#51, #52, #53, #56, #57, #58, #61, #62, #63, #66, #67, #68, #69
op	Operators & execute	#30, #34, #45, #46, #59, #60, #64, #65, #70
fn	Functions	#8, #9, #12, #13, #16, #17, #23, #24, #25, #27, #28, #29, #31, #32, #33, #35, #36, #37, #38, #39, #40, #41, #42, #43, #44, #50
mem	Memory, variables & registers	#1, #2, #3, #4, #5, #6, #7, #14, #47, #48

clr	**Clear, delete & power**	**#26, #54, #55**
mod	**Shift, modes, menus & navigation**	**#10, #11, #15, #18, #19, #20, #21, #22, #49**

2 · Features

49 features: 35 live on the keys, 9 app/UI features, 5 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **Four matrices MatA–MatD (up to 6×6) with a cell-by-cell editor**
2. LIVE **Three vectors VctA–VctC (up to 6 elements)**
3. LIVE **Literal matrix entry with [, ;] e.g. [1,2;3,4]**
4. LIVE **Matrix addition, subtraction and scalar multiplication**
5. LIVE **Matrix product A×B with dimension check**
6. LIVE **Element-wise (Hadamard) product ∘**
7. LIVE **Integer matrix powers A^n**
8. LIVE **Inverse A^{-1}**
9. LIVE **Determinant det(A)**
10. LIVE **Transpose Trn(A)**
11. LIVE **Reduced row-echelon form Rref(A) (Gauss-Jordan)**
12. LIVE **Rank(A)**
13. LIVE **Trace(A)**
14. LIVE **Eigenvalues Eig(A)**
15. LIVE **Adjugate Adj(A) and cofactor matrix Cof(A)**
16. LIVE **Identity matrix Ident(n)**
17. LIVE **Solve A·x = b with Solve(A,b)**
18. LIVE **Guided 2×2 / 3×3 linear-system solver form**
19. LIVE **LUP decomposition viewer (L, U and permutation)**
20. LIVE **Kronecker product Kron(A,B)**

21. LIVE Dimensions Dim(A)
22. LIVE Diagonal Diag(v or A)
23. LIVE Random integer matrix RanMat(r,c)
24. LIVE Dot product Dot(a,b)
25. LIVE Cross product Cross(a,b)
26. LIVE Vector length Norm(v)
27. LIVE Angle between vectors Angle(a,b) in the current unit
28. LIVE Unit vector Unit(v)
29. LIVE Projection Proj(a,b)
30. LIVE MatAns: every matrix result is kept for the next calculation
31. LIVE Matrix results drawn as a grid on the display
32. LIVE Matrix list screen showing all sizes and contents
33. LIVE Scalar calculator with trig, logs, powers and memory
34. LIVE History recall with ▲ ▼
35. LIVE SETUP menu with angle units and number formats
36. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
37. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
38. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
39. APP/UI Show / hide key numbers with one button (remembered between sessions)
40. APP/UI Copy the current display text to the clipboard

41. APP/UI **Physical-keyboard shortcuts: digits, + − * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i**
42. APP/UI **Fit / 1:1 zoom that never pushes keys off a phone screen**
43. APP/UI **Reference photo viewer and tool-tip name on every key**
44. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
45. PLANNED **Exam mode (locks memory & shows indicator)**
46. PLANNED **Answer history export to text**
47. PLANNED **Speech read-out of results**
48. PLANNED **Colour-blind safe key theme**
49. PLANNED **Custom shortcut keys**

2b · Feature guide

Calculator #25 is a algebraic matrix entry (MatA…MatD, VctA…VctC or [1,2;3,4], then EXE). Each card below is one of its 35 working features, with the keys that reach it and, where one exists, a verified example.

F1 Four matrices MatA–MatD (up to 6×6) with a cell-by-cell editor

Where: #1 MatA Recall MatA; #4 MatD Recall MatD; #9 LIST Show all matrices and vectors

See it in action: Determinant of MatA — *det(MatA), MatA = [[2,1],[1,3]]* = 5; Trace of a matrix product — *Trace(MatA×MatB)* = 19

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

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F2 Three vectors VctA–VctC (up to 6 elements)

Where: #5 VctA Recall VctA; #7 VctC Recall VctC; #9 LIST Show all matrices and vectors

See it in action: Dot product of two vectors — *Dot(VctA,VctB) = [1,2,3]·[4,5,6]* = 32

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F3 Literal matrix entry with [, ;] e.g. [1,2;3,4]

Where: #27 [Open a matrix literal; #28] Close a matrix literal; #8 EDIT Edit a matrix or vector (size, then cells)

See it in action: Determinant of a stored matrix — *det(MatB), MatB = [[1,2],[3,4]]* = -2; Trace of a matrix product — *Trace(MatA×MatB)* = 19

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F4 Matrix addition, subtraction and scalar multiplication

Where: #8 EDIT Edit a matrix or vector (size, then cells); #10 MAT► Matrix function catalogue; #14 MAns Recall the last matrix result MatAns

See it in action: Determinant of a stored matrix — *det(MatB), MatB = [[1,2],[3,4]]* = -2; Trace of a matrix product — *Trace(MatA×MatB)* = 19

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F5 Matrix product A×B with dimension check

Where: #8 EDIT Edit a matrix or vector (size, then cells); #10 MAT▶ Matrix function catalogue; #14 MAns Recall the last matrix result MatAns

See it in action: Determinant of a stored matrix — *det(MatB), MatB = [[1,2],[3,4]]* = -2; Dot product of two vectors — *Dot(VctA,VctB) = [1,2,3]·[4,5,6]* = 32

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F6 Element-wise (Hadamard) product ∘

Where: #33 x^2 Square x^2; #31 Dot Dot product; #32 Cross Cross product

See it in action: Dot product of two vectors — *Dot(VctA,VctB) = [1,2,3]·[4,5,6]* = 32; Trace of a matrix product — *Trace(MatA×MatB)* = 19

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F7 Integer matrix powers A^n

Where: #38 Solve Solve A·x=b; #8 EDIT Edit a matrix or vector (size, then cells); #10 MAT▶ Matrix function catalogue

See it in action: Determinant of a stored matrix — *det(MatB), MatB = [[1,2],[3,4]]* = -2; Trace of a matrix product — *Trace(MatA×MatB)* = 19

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F8 Inverse A^{-1}

Where: #23 A^{-1} Inverse / reciprocal; #39 sin Sine (uses the current angle unit); #40 cos Cosine (current angle unit)

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F9 Determinant det(A)

Where: #16 det Determinant

See it in action: Determinant of a stored matrix — *det(MatB), MatB = [[1,2],[3,4]]* = -2; Determinant of MatA — *det(MatA), MatA = [[2,1],[1,3]]* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F10 Transpose Trn(A)

Where: #17 Trn Transpose

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F11 Reduced row-echelon form Rref(A) (Gauss-Jordan)

Where: #24 Rref Reduced row echelon form; #29 ; Next matrix row

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F12 Rank(A)

Where: #24 Rref Reduced row echelon form

See it in action: Rank of the 3×3 identity — *Rank(MatC)*= 3

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F13 Trace(A)

Where: #25 Eig Eigenvalues

See it in action: Trace of a matrix product — *Trace(MatA×MatB)*= 19

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F14 Eigenvalues Eig(A)

Where: #25 Eig Eigenvalues

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F15 Adjugate Adj(A) and cofactor matrix Cof(A)

Where: #17 Trn Transpose; #16 det Determinant; #8 EDIT Edit a matrix or vector (size, then cells)

See it in action: Determinant of a stored matrix — *det(MatB), MatB = [[1,2],[3,4]]* = -2; Trace of a matrix product — *Trace(MatA×MatB)* = 19

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F16 Identity matrix Ident(n)

Where: #37 Ident Identity matrix; #8 EDIT Edit a matrix or vector (size, then cells); #10 MAT▶ Matrix function catalogue

See it in action: Determinant of a stored matrix — *det(MatB), MatB = [[1,2],[3,4]]* = -2; Trace of a matrix product — *Trace(MatA×MatB)* = 19

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F17 Solve A·x = b with Solve(A,b)

Where: #12 SYS Solve a 2×2 or 3×3 linear system; #38 Solve Solve A·x=b

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F18 Guided 2×2 / 3×3 linear-system solver form

Where: #12 SYS Solve a 2×2 or 3×3 linear system; #24 Rref Reduced row echelon form

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F19 LUP decomposition viewer (L, U and permutation)

Where: #13 LUP LUP decomposition of MatAns (or MatA)

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F20 Kronecker product Kron(A,B)

Where: #44 Kron Kronecker product; #31 Dot Dot product; #32 Cross Cross product

See it in action: Dot product of two vectors — *Dot(VctA,VctB) = [1,2,3]·[4,5,6]* = 32; Trace of a matrix product — *Trace(MatA×MatB)* = 19

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F21 Dimensions Dim(A)

Where: #44 Kron Kronecker product

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F22 Diagonal Diag(v or A)

Where: #37 Ident Identity matrix

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F23 Random integer matrix RanMat(r,c)

Where: #38 Solve Solve A·x=b; #8 EDIT Edit a matrix or vector (size, then cells); #10 MAT▶ Matrix function catalogue

See it in action: Determinant of a stored matrix — *det(MatB), MatB = [[1,2],[3,4]]* = -2; Trace of a matrix product — *Trace(MatA×MatB)* = 19

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F24 Dot product Dot(a,b)

Where: #31 Dot Dot product; #32 Cross Cross product; #33 x^2 Square x^2

See it in action: Dot product of two vectors — *Dot(VctA,VctB) = [1,2,3]·[4,5,6]* = 32; Trace of a matrix product — *Trace(MatA×MatB)* = 19

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F25 Cross product Cross(a,b)

Where: #32 Cross Cross product; #31 Dot Dot product; #33 x^2 Square x^2

See it in action: Dot product of two vectors — *Dot(VctA,VctB) = [1,2,3]·[4,5,6]* = 32; Trace of a matrix product — *Trace(MatA×MatB)* = 19

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F26 Vector length Norm(v)

Where: #36 Norm Vector length; #8 EDIT Edit a matrix or vector (size, then cells); #11 VCT▶ Vector function catalogue

See it in action: Length of a vector — *Norm([3,4])* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F27 Angle between vectors Angle(a,b) in the current unit

Where: #32 Cross Cross product; #39 sin Sine (uses the current angle unit); #40 cos Cosine (current angle unit)

See it in action: Dot product of two vectors — *Dot(VctA,VctB) = [1,2,3]·[4,5,6]* = 32

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F28 Unit vector Unit(v)

Where: #36 Norm Vector length; #22 MODE SETUP — angle unit, number format, display options; #39 sin Sine (uses the current angle unit)

See it in action: Length of a vector — *Norm([3,4])* = 5

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F29 Projection Proj(a,b)

Where: #31 Dot Dot product

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F30 MatAns: every matrix result is kept for the next calculation

Where: #14 MAns Recall the last matrix result MatAns; #29 ; Next matrix row; #8 EDIT Edit a matrix or vector (size, then cells)

See it in action: Determinant of a stored matrix — *det(MatB), MatB = [[1,2],[3,4]]* = -2; Trace of a matrix product — *Trace(MatA×MatB)* = 19

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F31 Matrix results drawn as a grid on the display

Where: #8 EDIT Edit a matrix or vector (size, then cells); #10 MAT▶ Matrix function catalogue; #14 MAns Recall the last matrix result MatAns

See it in action: Determinant of a stored matrix — *det(MatB), MatB = [[1,2],[3,4]]* = -2; Trace of a matrix product — *Trace(MatA×MatB)* = 19

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F32 Matrix list screen showing all sizes and contents

Where: #8 EDIT Edit a matrix or vector (size, then cells); #9 LIST Show all matrices and vectors; #10 MAT▶ Matrix function catalogue

See it in action: Determinant of a stored matrix — *det(MatB), MatB = [[1,2],[3,4]]* = -2; Trace of a matrix product — *Trace(MatA×MatB)* = 19

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F33 Scalar calculator with trig, logs, powers and memory

Where: #47 STO STO — store the value into a memory / variable (press the target next); #48 RCL RCL — recall a memory / variable

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F34 History recall with ▲ ▼

Where: #1 MatA Recall MatA; #2 MatB Recall MatB; #3 MatC Recall MatC

Try it: Press Reset, use the key(s) above, and write the display here:

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Your notes:

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F35 SETUP menu with angle units and number formats

Where: #22 MODE SETUP — angle unit, number format, display options; #32 Cross Cross product; #39 sin Sine (uses the current angle unit)

Try it: Press Reset, use the key(s) above, and write the display here:

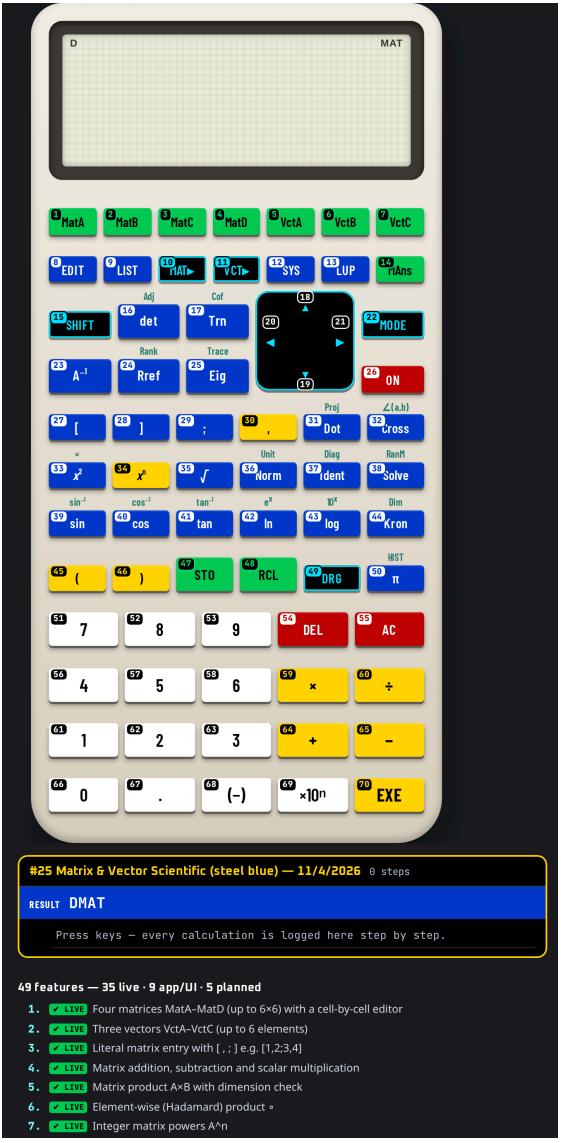
_____

Your notes:

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3 · Numbered key map



Calculator #25 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	MatA	·	·	·	mem
2	MatB	·	·	·	mem
3	MatC	·	·	·	mem
4	MatD	·	·	·	mem
5	VctA	·	·	·	mem
6	VctB	·	·	·	mem
7	VctC	·	·	·	mem
8	EDIT	·	·	·	fn
9	LIST	·	·	·	fn
10	MAT▶	·	·	·	mod
11	VCT▶	·	·	·	mod
12	SYS	·	·	·	fn
13	LUP	·	·	·	fn
14	MAns	·	·	·	mem
15	SHIFT	·	·	·	mod
16	det	Adj	·	·	fn
17	Trn	Cof	·	·	fn
18	▲	·	·	·	mod
19	▼	·	·	·	mod
20	◀	·	·	·	mod
21	▶	·	·	·	mod
22	MODE	·	·	·	mod
23	A^{-1}	·	·	·	fn

#	Face	Shift	Alpha	3rd	Group
24	Rref	Rank	·	·	fn
25	Eig	Trace	·	·	fn
26	ON	·	·	·	clr
27	[	·	·	·	fn
28	]	·	·	·	fn
29	;	·	·	·	fn
30	,	·	·	·	op
31	Dot	Proj	·	·	fn
32	Cross	∠(a,b)	·	·	fn
33	x^2	∘	·	·	fn
34	xn	·	·	·	op
35	√	·	·	·	fn
36	Norm	Unit	·	·	fn
37	Ident	Diag	·	·	fn
38	Solve	RanM	·	·	fn
39	sin	$\sin^{-1}$	·	·	fn
40	cos	$\cos^{-1}$	·	·	fn
41	tan	$\tan^{-1}$	·	·	fn
42	ln	ex	·	·	fn
43	log	10x	·	·	fn
44	Kron	Dim	·	·	fn
45	(	·	·	·	op
46	)	·	·	·	op
47	STO	·	·	·	mem
48	RCL	·	·	·	mem
49	DRG	·	·	·	mod
50	π	HIST	·	·	fn
51	7	·	·	·	num

#	Face	Shift	Alpha	3rd	Group
52	8	·	·	·	num
53	9	·	·	·	num
54	DEL	·	·	·	clr
55	AC	·	·	·	clr
56	4	·	·	·	num
57	5	·	·	·	num
58	6	·	·	·	num
59	×	·	·	·	op
60	÷	·	·	·	op
61	1	·	·	·	num
62	2	·	·	·	num
63	3	·	·	·	num
64	+	·	·	·	op
65	−	·	·	·	op
66	0	·	·	·	num
67	.	·	·	·	num
68	(−)	·	·	·	num
69	×10^{n}	·	·	·	num
70	EXE	·	·	·	op

4 · Key-by-key reference

Every key on calculator #25, in number order, with every function it carries in this app.

#1 MatA `Memory, variables & registers`

Primary: Recall MatA

#2 MatB `Memory, variables & registers`

Primary: Recall MatB

#3 MatC `Memory, variables & registers`

Primary: Recall MatC

#4 MatD `Memory, variables & registers`

Primary: Recall MatD

#5 VctA `Memory, variables & registers`

Primary: Recall VctA

#6 VctB `Memory, variables & registers`

Primary: Recall VctB

#7 VctC `Memory, variables & registers`

Primary: Recall VctC

#8 EDIT `Functions`

Primary: Edit a matrix or vector (size, then cells)

#9 LIST `Functions`

Primary: Show all matrices and vectors

#10 MAT▶ `Shift, modes, menus & navigation`

Primary: Matrix function catalogue

#11 VCT▶ `Shift, modes, menus & navigation`

Primary: Vector function catalogue

#12 SYS `Functions`

Primary: Solve a 2×2 or 3×3 linear system

#13 LUP `Functions`

Primary: LUP decomposition of MatAns (or MatA)

#14 MAns `Memory, variables & registers`

Primary: Recall the last matrix result MatAns

#15 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#16 det `Functions`

Primary: Determinant

shifted (2nd / f / SHIFT layer) (press #15 first): Adj — Adjugate

#17 Trn `Functions`

Primary: Transpose

shifted (2nd / f / SHIFT layer) (press #15 first): Cof — Cofactor matrix

#18 ▲ `Shift, modes, menus & navigation`

Primary: Cursor ▲ — up / previous history line / previous item

#19 ▼ `Shift, modes, menus & navigation`

Primary: Cursor ▼ — down / next history line / next item

#20 ◀ `Shift, modes, menus & navigation`

Primary: Cursor ◀ — move left in the line

#21 ▶ Shift, modes, menus & navigation

Primary: Cursor ▶ — move right in the line

#22 MODE Shift, modes, menus & navigation

Primary: SETUP — angle unit, number format, display options

#23 A^{-1} Functions

Primary: Inverse / reciprocal

Keyboard: i

#24 Rref Functions

Primary: Reduced row echelon form

shifted (2nd / f / SHIFT layer) (press #15 first): Rank — Rank

#25 Eig Functions

Primary: Eigenvalues

shifted (2nd / f / SHIFT layer) (press #15 first): Trace — Trace

#26 ON Clear, delete & power

Primary: ON — power on

#27 [Functions

Primary: Open a matrix literal

#28] Functions

Primary: Close a matrix literal

#29 ; Functions

Primary: Next matrix row

#30 , Operators & execute

Primary: Comma / argument separator (between function arguments, list items or coordinates)

Keyboard: ,

#31 Dot Functions

Primary: Dot product

shifted (2nd / f / SHIFT layer) (press #15 first): Proj — Projection of a on b

#32 Cross Functions

Primary: Cross product

shifted (2nd / f / SHIFT layer) (press #15 first): ∠(a,b) — Angle between vectors

#33 x^2 Functions

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #15 first): ∘ — Element-wise product

Keyboard: q

#34 xn Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

Keyboard: ^

#35 √ Functions

Primary: Square root √

Keyboard: r

#36 Norm Functions

Primary: Vector length

shifted (2nd / f / SHIFT layer) (press #15 first): Unit — Unit vector

#37 Ident Functions

Primary: Identity matrix

shifted (2nd / f / SHIFT layer) (press #15 first): Diag — Diagonal

#38 Solve Functions

Primary: Solve A·x=b

shifted (2nd / f / SHIFT layer) (press #15 first): RanM — Random integer matrix

#39 sin Functions

Primary: Sine (uses the current angle unit)

shifted (2nd / f / SHIFT layer) (press #15 first): $\sin^{-1}$ — Inverse sine $\sin^{-1}$

Keyboard: s

#40 cos Functions

Primary: Cosine (current angle unit)

shifted (2nd / f / SHIFT layer) (press #15 first): $\cos^{-1}$ — Inverse cosine $\cos^{-1}$

Keyboard: c

#41 tan Functions

Primary: Tangent (current angle unit)

shifted (2nd / f / SHIFT layer) (press #15 first): $\tan^{-1}$ — Inverse tangent $\tan^{-1}$

Keyboard: t

#42 ln Functions

Primary: Natural logarithm ln

shifted (2nd / f / SHIFT layer) (press #15 first): ex — e^x — exponential

Keyboard: n

#43 log Functions

Primary: Common logarithm $\log_{10}$

shifted (2nd / f / SHIFT layer) (press #15 first): 10x — 10^x — antilog

Keyboard: l

#44 Kron Functions

Primary: Kronecker product

shifted (2nd / f / SHIFT layer) (press #15 first): Dim — Dimensions

#45 (Operators & execute

Primary: Opening bracket (

Keyboard: (

#46) Operators & execute

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

Keyboard:)

#47 STO Memory, variables & registers

Primary: STO — store the value into a memory / variable (press the target next)

#48 RCL Memory, variables & registers

Primary: RCL — recall a memory / variable

#49 DRG `Shift, modes, menus & navigation`

Primary: DRG — choose/convert the angle unit (Deg / Rad / Grad)

#50 π `Functions`

Primary: π (3.14159265…) as an exact constant

shifted (2nd / f / SHIFT layer) (press #15 first): HIST — history

Keyboard: p

#51 7 `Digits & number entry`

Primary: Digit 7

Keyboard: 7

#52 8 `Digits & number entry`

Primary: Digit 8

Keyboard: 8

#53 9 `Digits & number entry`

Primary: Digit 9

Keyboard: 9

#54 DEL `Clear, delete & power`

Primary: DEL — delete the character at the cursor

Keyboard: Backspace Delete

#55 AC `Clear, delete & power`

Primary: AC — clear all of the current calculation

Keyboard: Escape

#56 4 `Digits & number entry`

Primary: Digit 4

Keyboard: 4

#57 5 `Digits & number entry`

Primary: Digit 5

Keyboard: 5

#58 6 `Digits & number entry`

Primary: Digit 6

Keyboard: 6

#59 × `Operators & execute`

Primary: Multiplication ×

Keyboard: * x

#60 ÷ `Operators & execute`

Primary: Division ÷

Keyboard: /

#61 **1** `Digits & number entry`

Primary: Digit 1

Keyboard: `1`

#62 **2** `Digits & number entry`

Primary: Digit 2

Keyboard: `2`

#63 **3** `Digits & number entry`

Primary: Digit 3

Keyboard: `3`

#64 **+** `Operators & execute`

Primary: Addition +

Keyboard: `+`

#65 **–** `Operators & execute`

Primary: Subtraction –

Keyboard: `-`

#66 **0** `Digits & number entry`

Primary: Digit 0

Keyboard: `0`

#67 . Digits & number entry

Primary: Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored

Keyboard: .

#68 (–) Digits & number entry

Primary: Negative sign (–) — makes the next number negative. Different from the subtraction key

#69 ×10ⁿ Digits & number entry

Primary: ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23

Keyboard: e

#70 EXE Operators & execute

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Keyboard: Enter =

5 · Worked examples

7 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Determinant of a stored matrix

Calculate **det(MatB), MatB = [[1,2],[3,4]]**. Press **Reset** first.

`#16` det → `#2` MatB → `#46`) → `#70` EXE

Display shows `-2` — expected -2. ✔ verified on the app.

Example 5.2 — Determinant of MatA

Calculate **det(MatA), MatA = [[2,1],[1,3]]**. Press **Reset** first.

`#16` det → `#1` MatA → `#46`) → `#70` EXE

Display shows `5` — expected 5. ✔ verified on the app.

Example 5.3 — Dot product of two vectors

Calculate **Dot(VctA,VctB) = [1,2,3]·[4,5,6]**. Press **Reset** first.

`#31` Dot → `#5` VctA → `#30` , → `#6` VctB → `#46`) → `#70` EXE

Display shows `32` — expected 32. ✔ verified on the app.

Example 5.4 — Trace of a matrix product

Calculate **Trace(MatA×MatB)**. Press **Reset** first.

`#15` SHIFT → `#25` Trace → `#1` MatA → `#59` × → `#2` MatB → `#46`) → `#70` EXE

Display shows `19` — expected 19. ✔ verified on the app.

Example 5.5 — Rank of the 3×3 identity

Calculate **Rank(MatC)**. Press **Reset** first.

`#15` SHIFT → `#24` Rank → `#3` MatC → `#46`) → `#70` EXE

Display shows `3` — expected 3. ✔ verified on the app.

Example 5.6 — Length of a vector

Calculate **Norm([3,4])**. Press **Reset** first.

`#36` Norm → `#27` [→ `#63` 3 → `#30` , → `#56` 4 → `#28`] → `#46`) → `#70` EXE

Display shows `5` — expected 5. ✔ verified on the app.

Example 5.7 — Determinant of a typed matrix

Calculate det([4,3;6,3]). Press Reset first.

#16 det → #27 [→ #56 4 → #30 , → #63 3 → #29 ; → #58 6 → #30 , → #63 3 → #28] → #46) → #70 EXE

Display shows -6 **— expected -6. ✔ verified on the app.**

6 · Tricks & hacks

Hack 1: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 2: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = …

Hack 3: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 4: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 5: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 6: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 7: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 8: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 9: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 10: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 11: Hidden functions on this model

17 extra functions sit on shifted layers. Press the layer key once, then the key: for example #15 then #16 → Adj; #15 then #17 → Cof; #15 then #24 → Rank; #15 then #25 → Trace.

Hack 12: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #70.

Hack 13: Type faster on a computer

32 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 14: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 15: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 16: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
i	#23	A^{-1}	Inverse / reciprocal
,	#30	,	Comma / argument separator (between function arguments, list items or coordinates)
q	#33	x^2	Square x^2
^	#34	xn	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
r	#35	√	Square root √
s	#39	sin	Sine (uses the current angle unit)
c	#40	cos	Cosine (current angle unit)
t	#41	tan	Tangent (current angle unit)
n	#42	ln	Natural logarithm ln
l	#43	log	Common logarithm $\log_{10}$
(	#45	(	Opening bracket (
)	#46	)	Closing bracket). Closing brackets at the end of a line may be omitted
p	#50	π	π (3.14159265…) as an exact constant
7	#51	7	Digit 7
8	#52	8	Digit 8
9	#53	9	Digit 9
Backspace Delete	#54	DEL	DEL — delete the character at the cursor
Escape	#55	AC	AC — clear all of the current calculation
4	#56	4	Digit 4
5	#57	5	Digit 5

6	#58	6	Digit 6
* x	#59	×	Multiplication ×
/	#60	÷	Division ÷
1	#61	1	Digit 1
2	#62	2	Digit 2
3	#63	3	Digit 3
+	#64	+	Addition +
-	#65	−	Subtraction −
0	#66	0	Digit 0
.	#67	.	Decimal point. Pressing it first gives "0."; a second point in the same number is ignored
e	#69	×10^{n}	×10^{x} / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^{x} 23
Enter =	#70	EXE	Execute / = — evaluate the line and show the result (EXE / ENTER / =)

App shortcuts

"# On / # Off"	show / hide key numbers
"Classic / Hi-Contrast"	switch key colours
"Copy"	copy display text
"Fit / 1:1"	scale to screen / design size
"Reset"	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 2: Wrong angle unit

If sin 30 gives –0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 3: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 4: Division precedence

$1 \div 2\pi$ may be read as $1 \div (2\pi)$ on textbook models. Type $1 \div (2 \times \pi)$ explicitly to be sure.

Mistake 5: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 6: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 7: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 8: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

Mistake 9: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

9 · Practice drills

Set A — Calculation drills (7)

Work each one on calculator #25, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. **det(MatB), MatB = [[1,2],[3,4]] =** _____
2. **det(MatA), MatA = [[2,1],[1,3]] =** _____
3. **Dot(VctA,VctB) = [1,2,3]·[4,5,6] =** _____
4. **Trace(MatA×MatB) =** _____
5. **Rank(MatC) =** _____
6. **Norm([3,4]) =** _____
7. **det([4,3;6,3]) =** _____

Set B — Key-finding drills (77)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. **Which key is "MatA" (Recall MatA)?** _____
2. **Which key is "MatB" (Recall MatB)?** _____
3. **Which key is "MatC" (Recall MatC)?** _____
4. **Which key is "MatD" (Recall MatD)?** _____
5. **Which key is "VctA" (Recall VctA)?** _____
6. **Which key is "VctB" (Recall VctB)?** _____
7. **Which key is "VctC" (Recall VctC)?** _____

8. Which key is “EDIT” (Edit a matrix or vector)? _____

9. Which key is “LIST” (Show all matrices and vectors)? _____

10. Which key is “MAT►” (Matrix function catalogue)? _____

11. Which key is “VCT►” (Vector function catalogue)? _____

12. Which key is “SYS” (Solve a 2×2 or 3×3 linear system)? _____

13. Which key is “LUP” (LUP decomposition of MatAns)? _____

14. Which key is “MAns” (Recall the last matrix result MatAns)? _____

15. Which key is “SHIFT” (SHIFT / 2nd)? _____

16. Which key is “det” (Determinant)? _____

17. Which key is “Trn” (Transpose)? _____

18. Which key is “ ▲ ” (Cursor ▲)? _____

19. Which key is “ ▼ ” (Cursor ▼)? _____

20. Which key is “◀” (Cursor ◀)? _____

21. Which key is “▶” (Cursor ▶)? _____

22. Which key is “MODE” (SETUP)? _____

23. Which key is “A^{-1}” (Inverse / reciprocal)? _____

24. Which key is “Rref” (Reduced row echelon form)? _____

25. Which key is “Eig” (Eigenvalues)? _____

26. Which key is “ON” (ON)? _____

27. Which key is “[” (Open a matrix literal)? _____

28. Which key is "]" (Close a matrix literal)? _____

29. Which key is ";" (Next matrix row)? _____

30. Which key is "," (Comma / argument separator)? _____

31. Which key is "Dot" (Dot product)? _____

32. Which key is "Cross" (Cross product)? _____

33. Which key is "x^2" (Square x^2)? _____

34. Which key is "xn" (Power x^y / ^)? _____

35. Which key is "√" (Square root √)? _____

36. Which key is "Norm" (Vector length)? _____

37. Which key is "Ident" (Identity matrix)? _____

38. Which key is "Solve" (Solve A·x=b)? _____

39. Which key is "sin" (Sine)? _____

40. Which key is "cos" (Cosine)? _____

41. Which key is "tan" (Tangent)? _____

42. Which key is "ln" (Natural logarithm ln)? _____

43. Which key is "log" (Common logarithm $\log_{10}$)? _____

44. Which key is "Kron" (Kronecker product)? _____

45. Which key is "(" (Opening bracket)? _____

46. Which key is ")" (Closing bracket). Closing brackets at the end of a line may be omitted)? _____

47. Which key is "STO" (STO)? _____

48. Which key is “RCL” (RCL)? _____

49. Which key is “DRG” (DRG)? _____

50. Which key is “π” (π)? _____

51. Which key is “DEL” (DEL)? _____

52. Which key is “AC” (AC)? _____

53. Which key is “×” (Multiplication ×)? _____

54. Which key is “÷” (Division ÷)? _____

55. Which key is “+” (Addition +)? _____

56. Which key is “−” (Subtraction −)? _____

57. Which key is “.” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

58. Which key is “(−)” (Negative sign)? _____

59. Which key is “$\times 10^{n}$” ($\times 10^{x}$ / EXP / EE)? _____

60. Which key is “EXE” (Execute / =)? _____

61. How do you reach “Adj”? _____

62. How do you reach “Cof”? _____

63. How do you reach “Rank”? _____

64. How do you reach “Trace”? _____

65. How do you reach “Proj”? _____

66. How do you reach “∠(a,b)”? _____

67. How do you reach “∘”? _____

68. How do you reach “Unit”? _____

69. How do you reach “Diag”? _____

70. How do you reach “RanM”? _____

71. How do you reach “$\sin^{-1}$”? _____

72. How do you reach “$\cos^{-1}$”? _____

73. How do you reach “$\tan^{-1}$”? _____

74. How do you reach “ex”? _____

75. How do you reach “10x”? _____

76. How do you reach “Dim”? _____

77. How do you reach “HIST”? _____

Appendix A — Drill answers

Set A

1. **-2** (display `-2`)
   `#16` det → `#2` MatB → `#46`) → `#70` EXE
2. **5** (display `5`)
   `#16` det → `#1` MatA → `#46`) → `#70` EXE
3. **32** (display `32`)
   `#31` Dot → `#5` VctA → `#30` , → `#6` VctB → `#46`) → `#70` EXE
4. **19** (display `19`)
   `#15` SHIFT → `#25` Trace → `#1` MatA → `#59` × → `#2` MatB → `#46`) → `#70` EXE
5. **3** (display `3`)
   `#15` SHIFT → `#24` Rank → `#3` MatC → `#46`) → `#70` EXE
6. **5** (display `5`)
   `#36` Norm → `#27` [→ `#63` 3 → `#30` , → `#56` 4 → `#28`] → `#46`) → `#70` EXE
7. **-6** (display `-6`)
   `#16` det → `#27` [→ `#56` 4 → `#30` , → `#63` 3 → `#29` ; → `#58` 6 → `#30` , → `#63` 3 → `#28`] → `#46`) → `#70` EXE

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #28
29. #29
30. #30
31. #31
32. #32
33. #33
34. #34
35. #35
36. #36
37. #37
38. #38
39. #39
40. #40
41. #41
42. #42
43. #43
44. #44
45. #45
46. #46
47. #47
48. #48
49. #49
50. #50
51. #54
52. #55
53. #59
54. #60
55. #64
56. #65
57. #67
58. #68
59. #69
60. #70
61. #15 SHIFT then #16 det
62. #15 SHIFT then #17 Trn
63. #15 SHIFT then #24 Rref
64. #15 SHIFT then #25 Eig
65. #15 SHIFT then #31 Dot
66. #15 SHIFT then #32 Cross
67. #15 SHIFT then #33 x^2
68. #15 SHIFT then #36 Norm
69. #15 SHIFT then #37 Ident
70. #15 SHIFT then #38 Solve
71. #15 SHIFT then #39 sin
72. #15 SHIFT then #40 cos
73. #15 SHIFT then #41 tan
74. #15 SHIFT then #42 ln
75. #15 SHIFT then #43 log
76. #15 SHIFT then #44 Kron
77. #15 SHIFT then #50 π

Appendix B — Function finder (A–Z)

Function	Label	Keys
×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23	×10ⁿ	#69
10ˣ — antilog	10x	#15 → #43
AC — clear all of the current calculation	AC	#55
Addition +	+	#64
Adjugate	Adj	#15 → #16
Angle between vectors	∠(a,b)	#15 → #32
Close a matrix literal	]	#28
Closing bracket). Closing brackets at the end of a line may be omitted	)	#46
Cofactor matrix	Cof	#15 → #17
Comma / argument separator (between function arguments, list items or coordinates)	,	#30
Common logarithm $\log_{10}$	log	#43
Cosine (current angle unit)	cos	#40
Cross product	Cross	#32
Cursor ▲ — up / previous history line / previous item	▲	#18
Cursor ▶ — move right in the line	▶	#21
Cursor ▼ — down / next history line / next item	▼	#19
Cursor ◀ — move left in the line	◀	#20
Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored	.	#67
DEL — delete the character at the cursor	DEL	#54
Determinant	det	#16

Diagonal	Diag	#15 → #37
Digit 0	0	#66
Digit 1	1	#61
Digit 2	2	#62
Digit 3	3	#63
Digit 4	4	#56
Digit 5	5	#57
Digit 6	6	#58
Digit 7	7	#51
Digit 8	8	#52
Digit 9	9	#53
Dimensions	Dim	#15 → #44
Division ÷	÷	#60
Dot product	Dot	#31
DRG — choose/convert the angle unit (Deg / Rad / Grad)	DRG	#49
Edit a matrix or vector (size, then cells)	EDIT	#8
Eigenvalues	Eig	#25
Element-wise product	∘	#15 → #33
e^x — exponential	ex	#15 → #42
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	EXE	#70
history	HIST	#15 → #50
Identity matrix	Ident	#37
Inverse / reciprocal	A^{-1}	#23
Inverse cosine $\cos^{-1}$	$\cos^{-1}$	#15 → #40

Inverse sine $\sin^{-1}$	$\sin^{-1}$	#15 → #39
Inverse tangent $\tan^{-1}$	$\tan^{-1}$	#15 → #41
Kronecker product	Kron	#44
LUP decomposition of MatAns (or MatA)	LUP	#13
Matrix function catalogue	MAT▸	#10
Multiplication ×	×	#59
Natural logarithm ln	ln	#42
Negative sign (−) — makes the next number negative. Different from the subtraction key	(−)	#68
Next matrix row	;	#29
ON — power on	ON	#26
Open a matrix literal	[	#27
Opening bracket (	(	#45
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	xn	#34
Projection of a on b	Proj	#15 → #31
Random integer matrix	RanM	#15 → #38
Rank	Rank	#15 → #24
RCL — recall a memory / variable	RCL	#48
Recall MatA	MatA	#1
Recall MatB	MatB	#2
Recall MatC	MatC	#3
Recall MatD	MatD	#4
Recall the last matrix result MatAns	MAns	#14
Recall VctA	VctA	#5
Recall VctB	VctB	#6

Recall VctC	VctC	#7
Reduced row echelon form	Rref	#24
SETUP — angle unit, number format, display options	MODE	#22
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	SHIFT	#15
Show all matrices and vectors	LIST	#9
Sine (uses the current angle unit)	sin	#39
Solve a 2×2 or 3×3 linear system	SYS	#12
Solve A·x=b	Solve	#38
Square root √	√	#35
Square x^2	x^2	#33
STO — store the value into a memory / variable (press the target next)	STO	#47
Subtraction –	–	#65
Tangent (current angle unit)	tan	#41
Trace	Trace	#15 → #25
Transpose	Trn	#17
Unit vector	Unit	#15 → #36
Vector function catalogue	VCT►	#11
Vector length	Norm	#36
π (3.14159265…) as an exact constant	π	#50

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

Matrix & vector scientific: four matrices and three vectors are kept between calculations, edited with a cell form, and combined in ordinary algebraic expressions.

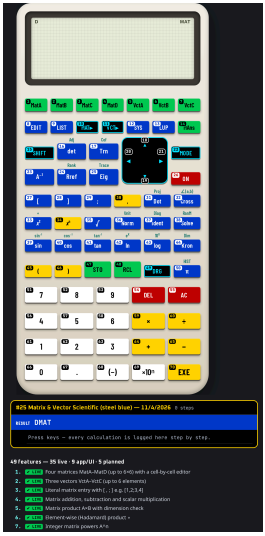
Rref, rank, cofactor and adjugate are U-45 code; determinant, inverse, eigenvalues and LU use the bundled math.js library (Apache-2.0).

Planned (not yet working)

- Exam mode (locks memory & shows indicator)
- Answer history export to text
- Speech read-out of results
- Colour-blind safe key theme
- Custom shortcut keys

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 MatA	#2 MatB	#3 MatC
#4 MatD	#5 VctA	#6 VctB
#7 VctC	#8 EDIT	#9 LIST
#10 MAT▶	#11 VCT▶	#12 SYS
#13 LUP	#14 MAns	#15 SHIFT
#16 det ⇧ Adj	#17 Trn ⇧ Cof	#18 ▲
#19 ▼	#20 ◀	#21 ▶
#22 MODE	#23 A^{-1}	#24 Rref ⇧ Rank
#25 Eig ⇧ Trace	#26 ON	#27 [
#28]	#29 ;	#30 ,
#31 Dot ⇧ Proj	#32 Cross ⇧ ∠(a,b)	#33 x^2 ⇧ ∘
#34 xn	#35 √	#36 Norm ⇧ Unit
#37 Ident ⇧ Diag	#38 Solve ⇧ RanM	#39 sin ⇧ $\sin^{-1}$
#40 cos ⇧ $\cos^{-1}$	#41 tan ⇧ $\tan^{-1}$	#42 ln ⇧ ex
#43 log ⇧ 10x	#44 Kron ⇧ Dim	#45 (
#46)	#47 STO	#48 RCL
#49 DRG	#50 π ⇧ HIST	#51 7

#52 8	**#53** 9	**#54** DEL
#55 AC	**#56** 4	**#57** 5
#58 6	**#59** ×	**#60** ÷
#61 1	**#62** 2	**#63** 3
#64 +	**#65** −	**#66** 0
#67 .	**#68** (−)	**#69** $\times 10^{n}$
#70 EXE		

U-45 CALCULATORS · Calculator #25 · cut out and keep